FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2024

<u>TEST DATE: 20-01-2024</u>

Time Allotted: 3 Hours **Maximum Marks: 300**

<u>General Instructions:</u>

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

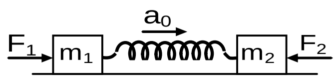
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two blocks m_1 and m_2 are connected with a compressed spring and placed on a smooth horizontal surface as shown in figure.



Force constant of spring is k. Under the influence of forces F_1 and F_2 at an instant blocks move with common acceleration a_0 At that instant force F_2 is suddenly withdrawn. Mark correct option.

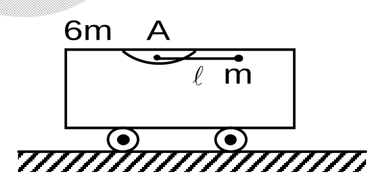
(A) Instantaneous acceleration of m_2 is $a_0 + \frac{F_1}{m_1}$

(B) Instantaneous acceleration of m_2 is $a_0 + \frac{F_2}{m_2}$

(C) Instantaneous acceleration of m_1 is $a_1 = 0$

(D) Instantaneous acceleration of m_2 is $a_2 = 0$

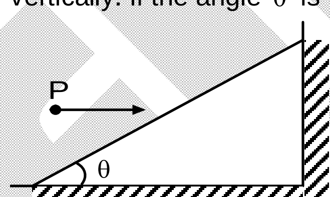
2. In the figure shown the cart of mass 6 m is initially at rest. A particle of mass m is attached to the end of the light rod which can rotate freely about A. If the rod is released form rest in a horizontal position shown, determine the velocity V_{rel} of the particle with respect to the cart when the rod is vertical. Assume friction less surface.



(A) $\sqrt{\frac{7}{3}gl}$ (B) $\sqrt{\frac{7}{6}gl}$

(C) $\sqrt{\frac{14}{3}gl}$ (D) $\sqrt{\frac{8}{3}gl}$

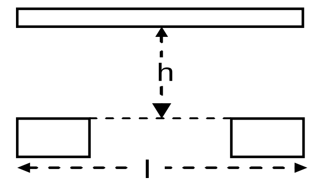
3. In the figure shown a particle P strikes the inclined fixed smooth plane horizontally and rebounds vertically. If the angle θ is 60^o, then the coefficient of restitution is:



(A) 1/3 (B) $1\sqrt{3}$

(C) 1/2 (D) 1

4. A uniform horizontal rod of length l falls vertically from height h on two identical blocks placed symmetrically below the rod as shown in figure. The coefficients of restitution are e_1 and e_2. The maximum height through which the centre of mass of the rod will rise after bouncing off the blocks is

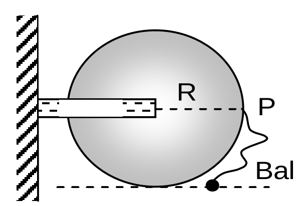


(A) $\frac{h}{(e_1+e_2)}$
(B) $\frac{(e_1+e_2)^2 h}{4}$
(C) $\frac{(e_1+e_2)^2 h}{2}$
(D) $\frac{4h}{(e_1^2+e_2^2)}$

5. The displacement of a particle is represented by the equation $y = \sin^3 \omega t$. The motion is
(A) Non – periodic
(B) Periodic but not simple harmonic
(C) Simple harmonic with period $2\pi/\omega$
(D) Simple harmonic with period π/ω

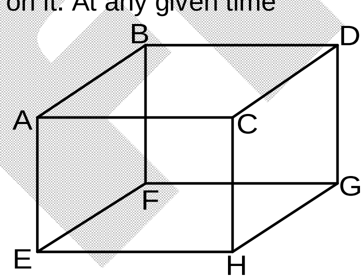
6. A steel ring of radius r and cross-section area A is fitted on to a wooden disc of radius $R(R > r)$. If Young's modulus be E, then the force with which the steel ring is expanded is:
(A) $AR\frac{R}{r}$
(B) $AE\left(\frac{R-r}{r}\right)$
(C) $\frac{E}{A}\left(\frac{R-r}{A}\right)$
(D) $\frac{Er}{AR}$

7. Figure shows a disc of mass M and radius R hinged at the centre. A small ball of mass $\frac{M}{2}$ is attached to point P with a thread of length 2R and held at rest at position shown. Now, the ball is released to fall under gravity with what angular speed the disc starts turning when the string becomes taut?



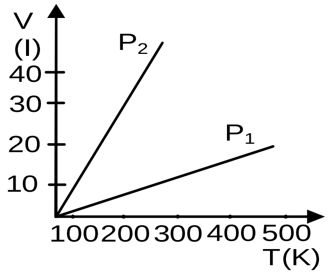
(A) $\sqrt{\frac{g}{2R}}$
(B) $\sqrt{\frac{g}{R}}$
(C) $\sqrt{\frac{R}{g}}$
(D) $\sqrt{\frac{2g}{R}}$

8. 1 mole of an ideal gas is contained in a cubical volume V, ABCDEFGH at 300 K in figure. One face of the cube (EFGH) is made up of a material which totally absorbs any gas molecule incident on it. At any given time



(A) The pressure on EFGH would be zero
(B) The pressure on all the faces will be equal
(C) The pressure of EFGH would be double the pressure on ABCD
(D) The pressure on EFGH would be half that on ABCD

9. Volume versus temperature graphs for a given mass of an ideal gas are shown in figure at two different values of constant pressure. What can be inferred about relation between P_1 & P_2?



(A) $P_1 > P_2$
(B) $P_1 = P_2$
(C) $P_1 < P_2$
(D) Data is insufficient

10. A positively charged particle is released from rest in an uniform electric field. The electric potential energy of the charge
(A) Remains a constant because the electric field is uniform
(B) Increases because the charge moves along the electric field
(C) Decreases because the charge moves along the electric field
(D) Decreases because the charge moves opposite to the electric field

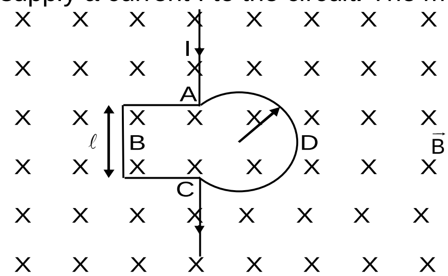
11. A hollow conducting sphere of inner radius R and outer radius 2R has resistivity 'ρ' a function of the distance 'r' from the centre of the sphere: $\rho = kr^2/R$. The inner and outer surfaces are painted with a perfectly conducting 'paint' and a potential difference ΔV is applied between the two surfaces. Then as 'r' increases from R to 2R, the electric field inside the sphere
(A) Increases
(B) Decreases
(C) Remains constant
(D) Passes through a maxima

12. A galvanometer shows a reading of 0.65 mA. When a galvanometer is shunted with a 4Ω resistance, the deflection is reduced to 0.13 mA. If the resistance of 2Ω is further connected parallel to 4Ω resistance (i.e. galvanometer is further shunted with a 2Ω wire) the new reading will be (Assume main current remains the same)
(A) 0.60 mA
(B) 0.8 mA
(C) 0.12 mA
(D) 0.05 mA

13. A soap bubble of radius R is blown. After heating the solution a second bubble of radius 2R is blown. The work required to blow the second bubble in comparison to that required for the first bubble is:
(A) Double
(B) Slightly less than double
(C) Slightly less than four times
(D) Slightly more than four times

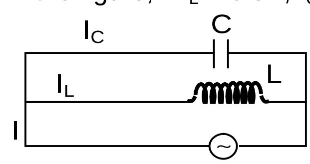
14. In an experiment to measure the length of a rod by four different instruments, the measurement is reported as
(i) 200.0 cm (ii) 20 cm (iii) 20.00 cm (iv) 0.200 cm
From the above data, we can infer that
(A) All measurements have same accuracy
(B) A has least accuracy
(C) B has least accuracy
(D) D has least accuracy

15. The figure shown a uniform conducting loop ABCDA placed in a uniform magnetic field perpendicular to its plane. The part ABC is the $(3/4)^{th}$ portion of the square of side length ℓ. The part ADC is a circular arc of radius R. The points A and C are connected to a battery which supply a current I to the circuit. The magnetic force on the loop due to the field B is



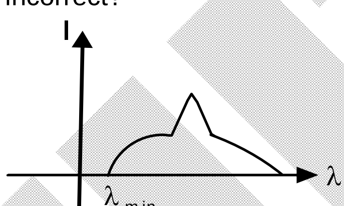
(A) Zero
(B) $BI\ell$
(C) 2BIR
(D) $\frac{BI\ell R}{I+R}$

16. In the figure, if I_L = 0.8A, I_C = 0.6A, then I = ? (given currents are rms values)



(A) 0.4 A
(B) 0.2 A
(C) 1.0 A
(D) 1.4 A

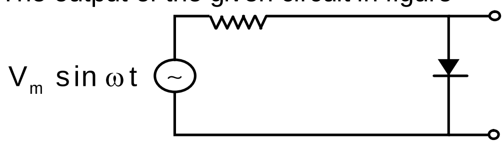
17. An x – ray tube has three main controls.
(i) The target material (its atomic number Z)
(ii) The filament current (I_f) and
(iii) The acceleration voltage (V)
Figure shows a typical intensity distribution against wavelength. Which of the following is incorrect?



(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X ray increases if V increased
(D) The intensity everywhere increases if filament current I_f is increased

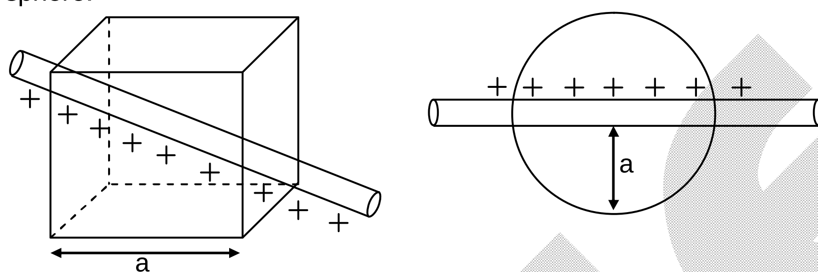
18. Which of the following statement is correct in connection with hydrogen spectrum?
(A) The longest wavelength in the Balmer series is longer than the longest wavelength in Lyman series
(B) The shortest wavelength in the Balmer series is shorter than the shortest wavelength in the Lyman series
(C) The longest wavelength in both Balmer and Lyman series are equal
(D) The longest wavelength in Balmer series is shorter than the longest wavelength in the Lyman series.

19. The output of the given circuit in figure



(A) Would be zero at all times
(B) Would be like a half wave rectifier with positive cycles output
(C) Would be like a half wave rectifier with negative cycles output
(D) Would be like that of a full wave rectifier

20. A linear charge having linear charge density λ penetrates a cube diagonally and then it penetrates a sphere diametrically as shown. What will be the ratio of flux coming cut of cube and sphere.



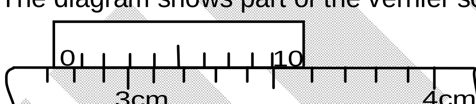
(A) $\frac{1}{2}$
(B) $\frac{2}{\sqrt{3}}$
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{1}{1}$

SECTION – B

(Numerical Answer Type)

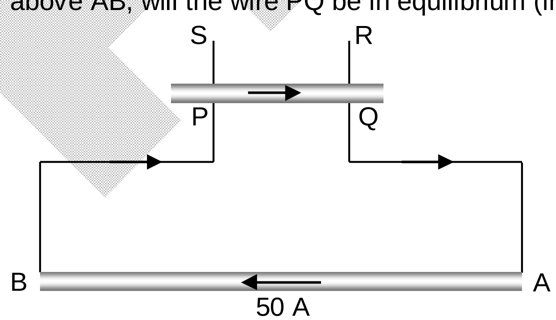
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. The diagram shows part of the vernier scale on a pair of calipers.

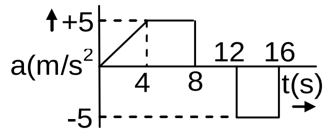


If the correct reading is "n" cm. Then the volume of "100n" is___

22. A long wire AB is placed on a table. Another wire PQ of mass 1.0 g and length 50 cm is set to slide on two rails PS and QR. A current of 50 A is passed through the wires. At what distance above AB, will the wire PQ be in equilibrium (in mm):



23. The acceleration of a train between two stations 2 km apart is shown in the figure. The maximum speed of the train is ____ (m/s)

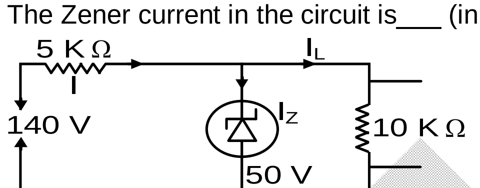


24. A ray of light incident at an angle θ on a refracting face of a prism emerges from the other face normally. If the angle of the prism is 5^o and the prism is made of a material of refractive index 1.5 the value of 10θ is (in degree)

25. If the light of wavelength of maximum intensity emitted from surface at temperature T_1 is used to cause photoelectric emission from a metallic surface, the maximum kinetic energy of the emitted electron is 6 eV, which is 3 times the work function of the metallic surface. If light of wavelength of maximum intensity emitted from a surface at temperature $T_2(T_2 = 2T_1)$ is used, the maximum kinetic energy of the photoelectrons emitted is

26. The Q – factor of a series LCR circuit with L = 3mH, C = 2.7μF, $R = \frac{10}{3}\Omega$,___

27. The Zener current in the circuit is___ (in mA)



28. One mole of ideal gas whose adiabatic exponent $\gamma = \frac{4}{3}$ undergoes process $P = 200 + \frac{1}{V}$, then change in internal energy of gas when volume changes from 2m^3 to 4m^3 is (in Joules)

29. A container filled with liquid of density $10^3 kg/m^2$ is kept on horizontal surface. An orifice of area 10 cm^2 is made in the wall of container at a distance 100 cm below the free surface of liquid. Find the thrust on the container at the instant when height of liquid is 50 cm above the orifice is ____(in newton's) (Assuming container is at rest and area of orifice is very small than the area of the container) Take g = 10 m/s^2

30. An object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and plane mirror is 30 cm, it is found that there is no parallax between the images formed by two mirrors. Radius of curvature of mirror will be: (in cm)

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. If we accelerate α–particle through a potential difference of V volts. The de-Broglie wave length associated with it is:

(A) $\sqrt{150.V^{-1}}\,A^o$
(B) $0.286.\,V^{-\frac{1}{2}}\,A^o$
(C) $\frac{0.101}{\sqrt{V}}A^o$
(D) $\frac{0.983}{\sqrt{V}}A^o$

32. A certain quantity of a gas occupies 100 mL when collected over water at $15\,^\circ C$ and 750 mm pressure. It occupies 91.9 mL in dry state at NTP. Find the aqueous vapour pressure at $15\,^\circ C$?

(A) 15.7 mm
(B) 7.9 mm
(C) 736.8 mm
(D) 13.2 mm

33. For the reversible reaction

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

At 500^0C, the value of K_P is 1.44×10^{-5} when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mole $litre^{-1}$, is

(A) $\frac{1.44\times10^{-5}}{(0.082\times500)^{-2}}$
(B) $\frac{1.44\times10^{-5}}{(8.314\times773)^{-2}}$
(C) $\frac{1.44\times10^{-5}}{(0.082\times773)^{2}}$
(D) $\frac{1.44\times10^{-5}}{(0.082\times773)^{-2}}$

34. Predict the order of $\Delta_\circ$ for the following compounds

I $\left[Mn(H_2O)_6\right]^{2+}$

II $\left[Mn(CN)_2(H_2O)_4\right]$

III $\left[Mn(CN)_4(H_2O)_2\right]^{2-}$

(A) $\Delta_\circ$ (I) < $\Delta_\circ$ (II) < $\Delta_\circ$ (III)
(B) $\Delta_\circ$ (II) < $\Delta_\circ$ (I) < $\Delta_\circ$ (III)
(C) $\Delta_\circ$ (III) < $\Delta_\circ$ (II) < $\Delta_\circ$ (I)
(D) $\Delta_\circ$ (I) < $\Delta_\circ$ (III) < $\Delta_\circ$ (II)

35. Match the items of List I and List II:

Reactions	**Unit of rate constant**
A. Decomposition of N_2O on hot Pt.	(I) $mol^{-2}L^2Time^{-2}$
B. Hydrolysis of ester by acid	(II) $mol\ L^{-1}Time^{-1}$
C. Alkaline hydrolysis of ester	(III) $mol^{-1}L\ Time^{-1}$
D. Reduction of $FeCl_3$ by $SnCl_2$	(IV) $Time^{-1}$

(A) A – III, B – II, C – I, D – IV
(B) A – I, B – II, C – III, D – IV
(C) A – IV, B – II, C – I, D – III
(D) A – II, B – IV, C – III, D – I

36. The oxidations states of X, Y and Z are +2, +5 and –2 respectively. The possible formula of compound formed by them is:

(A) XYZ_2
(B) $X_3(Y_4Z)_2$
(C) $Y_2(XZ_3)_2$
(D) $X_3(YZ_4)_2$

37. The IP_1, IP_2, IP_3, IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8 and 162.2 eV respectively. The element is likely to be

(A) Cl
(B) Na
(C) C
(D) Ba

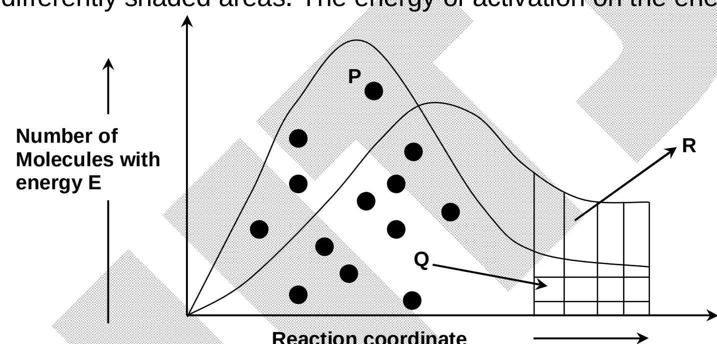
38. Which option is ***not*** matched in correct sequence?

(A) d^5, d^3, d^1, $d^4 \longrightarrow$ increasing magnetic moment
(B) MO, M_2O_3, MO_2, $M_2O_5 \longrightarrow$ decreasing basic strength
(C) Sc, V, Cr, Mn $\longrightarrow$ increasing number of oxidation states
(D) Co^{2+}, Fe^{+3}, Cr^{+3}, $Sc^{+3} \longrightarrow$ increasing stability

39. $K_2[HgI_4]$ detects the ion/group

(A) NH_2
(B) NO
(C) NH_4^+
(D) Cl^-

40. The distribution of the number of molecules with energy E is given in the figure for two temperatures T_1 and a higher temperature T_2. The letters P, Q, R refer to the separate and differently shaded areas. The energy of activation on the energy axis



Which expression gives the fraction of the molecules present which have at least activation energy at the higher temperature T_2.

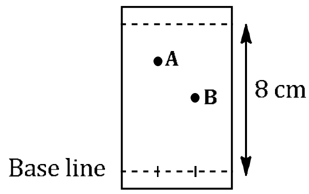
(A) $\frac{Q}{P}$
(B) $\frac{Q+R}{P+Q}$
(C) $\frac{Q+R}{P}$
(D) $\frac{Q+R}{P+Q-R}$

41. Arrange the following compounds in their decreasing basic strength in aq. Solution.

$CH_3-\overset{\overset{NH}{\|}}{C}-NH_2$ (I) $CH_3\overset{\overset{O}{\|}}{C}-NH_2$ (II) $CH_3CH_2NH_2$ (III) $(CH_3)_2NH$ (IV)

(A) IV > II > III > I
(B) II > I > IV > III
(C) I > IV > III > II
(D) IV > I >II > III

42. From the following TLC plate



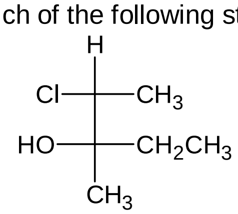
if the R_f values of A and B are 0.75 and 0.5 respectively, then the difference in the height of A and B on TLC plate is

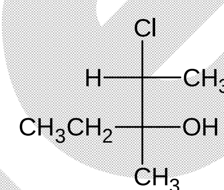
(A) 1 cm
(B) 1.5 cm
(C) 2 cm
(D) 2.5 cm

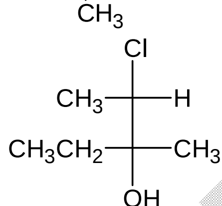
43. What is the order of reactivity of the alkenes $\left(CH_3\right)_2C = CH_2\,(I)$, $CH_3CH = CH_2\,(II)$ and $CH_2 = CH_2\,(III)$ when subject to acid-catalyzed hydration?

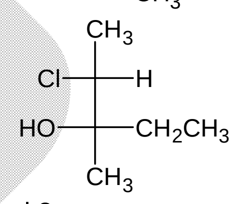
(A) I > II > III
(B) I > III > II
(C) III > II > I
(D) II > I > III

44. Which of the following structure are superimposable?

(1)


(2)


(3)


(4)


(A) 1 and 2
(B) 2 and 3
(C) 1 and 4
(D) 1 and 3

45. In Williamson synthesis of mixed ether having a primary alcohol and a tertiary alkyl group if tertiary halide is used then:

(A) Rate of reaction will be slow due to slow cleavage of carbon-halogen bond
(B) Alkene will be the main product
(C) Simple ether will form instead of mixed ether
(D) Expected mixed ether will be formed

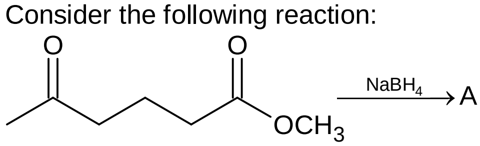
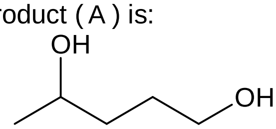
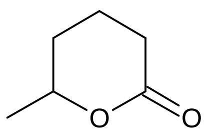
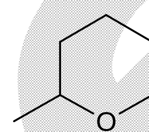
46. When HCl gas is passed through propene in the presence of benzoyl peroxide it gives

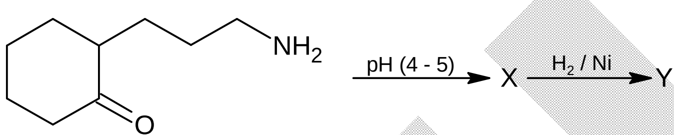
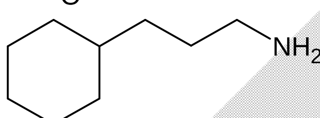
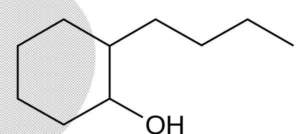
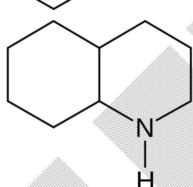
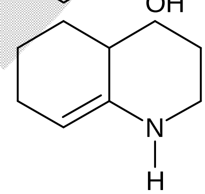
(A) n-propyl chloride
(B) 2-chloropropane
(C) allyl chloride
(D) no reaction

47. At 25^oC, pK_b of NH_2 and pK_a of COOH are x and y respectively. Then, pI of the glycine is

(A) $\dfrac{x+y}{2}$
(B) $\dfrac{14+x+y}{2}$
(C) $\dfrac{14+x-y}{2}$
(D) $\dfrac{14-x+y}{2}$

48. The formation constant of complex HgI_4^{2-} at 25°C is
Given: $E^\circ_{(Hg^{2+}/Hg)}$ = 0.85 V and $E^\circ_{(HgI_4^{2-}/Hg,\, I^-)}$ = – 0.05 V.
(A) $1.0\times10^{-26.66}$
(B) 1.0×10^{60}
(C) $1.0\times10^{26.66}$
(D) 1.0×10^{30}

49. Consider the following reaction:

The product (A) is:
(A) 
(B) 
(C) 
(D) 

50.  Here Y is:
(A) 
(B) 
(C) 
(D) 

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Find out the change in pH of water (at 25℃) when a drop of 1 M HCl is dipped in 1 litre H_2O. (Given 100 drop = 1 mL)

52.

Given Reaction	Energy Change (in kJ)
$Li(s)\rightarrow Li(g)$	161
$Li(g)\rightarrow Li^{+}(g)$	520
$\frac{1}{2}F_2(g)\rightarrow F(g)$	77
$F(g)+e^{-}\rightarrow F^{-}(g)$	(Electron gain enthalpy)

$Li^{+}(g)+F^{-}(g)\rightarrow Li\,F(s)$ – 1047

$Li(s)+\frac{1}{2}F_2(g)\rightarrow LiF(s)$ – 617

Based on data provided the value of electron gain enthalpy (in $kJ\,mol^{-1}$) of fluorine is (neglecting sign):

53. The pK_{a_1}, pK_{a_2} and pK_{a_3} values for the amino acid cysteine are respectively 1.8, 8.0, 10.0.

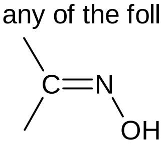
$$\left(HS-CH_2-\underset{\substack{|\\ NH_2}}{CH}-COOH\right)$$

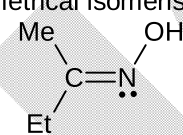
The isoelectric point of cysteine amino acid is

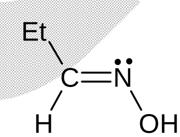
54. If formic acid, acetic acid, propanoic acid and benzoic acid is mixed with Phosphorus and Bromine then the number of products (including stereoisomer) formed are

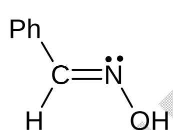
55. The number of 3d electrons occupied in t_{2g} orbitals of hydrated Cr^{3+} ion (octahedral).

56. How many of the following can show geometrical isomerism?

(I) 

(II) 

(III) 

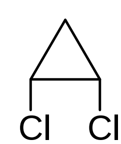
(IV) 

(V) 

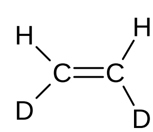
(VI) CHCl = C = CHCl

(VII) Ph – N = N – Ph

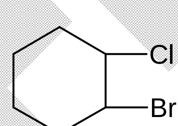
(VIII) MeCH = N – NH_2

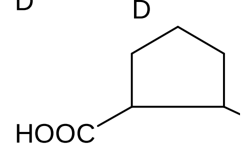
(IX) 

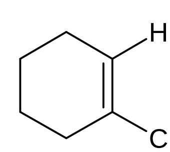
(X) CHCl = C = C = CHCl

(XI) 

(XII) $CH_3 - C \equiv C - CH_2Cl$

(XIII) 

(XIV) 

(XV) 

(XVI) 

57. For the strong electrolytes NaOH, NaCl and $BaCl_2$ the molar ionic conductivities at infinite dilution are 250, 125 and 300 $\text{mho cm}^2\,\text{mol}^{-1}$ respectively. The molar conductivity of $Ba(OH)_2$ at infinite dilution ($\text{mho cm}^2\,\text{mol}^{-1}$) is

58. Two moles of a gas at 8.21 bar and 300K are expand at constant temperature up to 2.73 bar against a constant pressure of 1 bar. How much work is done by the gas in atm-litre? (neglect the sign)

59. 30 ml of 0.2 M NaOH is added with 50 ml 0.2 M CH_3COOH solution. The extra volume of 0.2 M NaOH required to make the pH of the solution 5.00 is $\frac{10}{x}$ ml. The value of x is: (The ionization constant of $CH_3COOH = 2\times10^{-5}$)

60. The combustion of sodium in excess air yields a higher oxide. What is the oxidation state of the oxygen in the product? Neglect the negative sign.

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $y = \tan^{-1}\left(\frac{4x}{1+5x^2}\right) + \tan^{-1}\left(\frac{2+3x}{3-2x}\right)$, where $x \in \left(0, \frac{2}{3}\right)$. If $\frac{dy}{dx} = \frac{\alpha}{1+25x^2}$, then the value of α is equal to:
(A) 3
(B) 4
(C) 5
(D) 6

62. If the value of $\lim_{n\to\infty} \sum_{k=0}^{n} \frac{{}^nC_k}{n^k(k+3)}$ equals L. Then $[L]$ is equal to
[Note: Where [k] denotes greatest integer function less than or equal to k]
(A) 0
(B) 1
(C) 2
(D) 3

63. Let $f(x)$ and $g(x)$ are functions defined in the real domain and co – domain, such that $\sqrt{1-f^2(x)} = g(x)$, then which of the following statements are necessarily true?
(A) If g(x) is periodic with period 1, then f(x) is periodic with period half.
(B) If f'(c) = - f(c) = 0.5, then $\frac{g'(c)}{g(c)} = \frac{1}{3}$
(C) If g(x) is an even function, then f(x) is odd.
(D) If g(x) is continuous function then f(x) is also continuous in their respective domains.

64. The value of $\lim_{x\to\infty} \frac{e^x\left(\left(2^{x^n}\right)^{1/e^x} - \left(3^{x^n}\right)^{1/e^x}\right)}{x^n}$ where n is positive integer, is:
(A) In 2 – In 3
(B) In 3 – In 2
(C) 0
(D) None of these

65. Let f be an invertible function form $R \to R$ satisfying the equation $f^3(x) - (x^3+2)f^2(x) + (2x^3+1)f(x) - x^3 = 0$. Then the value of $f'(8) \times (f^{-1})'(8)$, is:
(A) 12
(B) 16
(C) 20
(D) 32

66. If $\int \frac{3\tan\left(x-\frac{\pi}{4}\right)}{\cos^2 x\sqrt{\tan^3 x + \tan^2 x + \tan x}}dx = k\tan^{-1}\left(\sqrt{\tan x + 1 + \cot x}\right) + C$, then the value of k is: [where C is constant of integration]
(A) 2
(B) 3
(C) 6
(D) 8

67. The solution of the differential equation $e^{-x}(y+1)dy+(\cos^2 x-\sin 2x)y\,dx=0$ subjected to condition that y = 1 when x = 0, is:
(A) $(y+1)+e^x\cos^2 x=2$
(B) $y+\ln y=e^x\cos^2 x$
(C) $\ln(y+1)+e^x\cos^2 x=1$
(D) $y+\ln y+e^x\cos^2 x=2$

68. Let f be a function defined by $y=f(x)$ where $x=2t-|t|$ and $y=t^2+t|t|$ for $t\in R$, then:
(A) f(x) is both continuous and differentiable at x = 0
(B) f(x) is non differentiable at x = 0
(C) f(x) is discontinuous at x = 0
(D) f(x) is neither continuous nor differentiable at x = 0

69. If $32\tan^8\theta=2\cos^2\alpha-3\cos\alpha$ and $3\cos 2\theta=1$, then the most general value of α are:
(A) $n\pi-\frac{\pi}{3}$
(B) $2n\pi\pm\frac{2\pi}{3}$
(C) $n\pi+\frac{\pi}{3}$
(D) $n\pi\pm\frac{\pi}{3}$
(Where $n\in I$)

70. The difference between the radii of the largest and the smallest circles, which have their centres on the circumference of the circle $x^2+2x+y^2+4y=4$, and pass through the point (3, 4) is equal to:
(A) 4
(B) 6
(C) 8
(D) 10

71. If a line with direction ratio 2:1:1 intersects the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{1}=\frac{y+1}{2}=\frac{z+2}{3}$ at A and B then $|\overline{AB}|$ is:
(A) $2\sqrt{3}$
(B) $2\sqrt{5}$
(C) $2\sqrt{6}$
(D) $2\sqrt{7}$

72. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}|=2|\vec{b}|=4|\vec{c}|$ and $2\vec{a}-3\vec{b}+6\vec{c}=\vec{0}$. If θ is the angle between $\vec{a}$ and $\vec{b}$, then $\cos\theta$ equals:
(A) $\frac{2}{3}$
(B) $\frac{1}{3}$
(C) $\frac{3}{5}$
(D) $\frac{4}{5}$

73. Let P and Q are points on the parabola $y^2=4ax$ with vertex O, such that OP is perpendicular to OQ and have length r_1 and r_2 respectively, then the value of $\frac{r_1^{4/3}r_2^{4/3}}{r_1^{2/3}+r_2^{2/3}}$ is:
(A) $16a^2$
(B) a^2
(C) $4a$
(D) None of these

74. Consider the ellipse $\frac{x^2}{f\left(k^2+2k+5\right)}+\frac{y^2}{f\left(k+11\right)}=1$, where f(x) is a positive decreasing function, then the value of k for which major axis coincides with x-axis is:
(A) $k\in(-7,\ -5)$
(B) $k\in(-5,\ -3)$
(C) $k\in(-3,\ 2)$
(D) None of these

75. Let A_1, A_2, A_3, ……, A_n are the vertices of a polygon of n sides. If the number of pentagons that can be constructed by joining these vertices such that none of the side of the polygon is also the side of the pentagon is 36, then the value of n is equal to:
(A) 8
(B) 10
(C) 11
(D) 12

76. Let $\left(1+3x+2x^2\right)^6=\sum_{k=0}^{12}a_kx^k$
(A) The coefficient of x^{12} equals 2^{12}
(B) The coefficient of x^{11} equals $9(2)^8$
(C) $\frac{\sum_{k=0}^{6}a_{2k}}{a_{12}}=\frac{3^6}{2}$
(D) $\frac{\sum_{k=1}^{6}a_{2k-1}}{a_{12}}=2(3)^6$

77. If α,β,γ are the roots of the equation $x^3-px+1=0,\ p\in R$ and A, B, C three matrices defined as:
$A=\begin{bmatrix}\alpha^3+1 & -2 & 3\\ 2 & \beta^3+1 & 1\\ -3 & -1 & \gamma^3+1\end{bmatrix}, B=\begin{bmatrix}\beta^3-2 & 3 & -1\\ -3 & \gamma^3-2 & -2\\ 1 & 2 & \alpha^3-2\end{bmatrix}$ and $C=\begin{bmatrix}\gamma^3+5 & -1 & -2\\ 1 & \alpha^3+5 & 1\\ 2 & -1 & \beta^3+5\end{bmatrix}$. Then the value of det. (A + B + C) equals:
(A) 0
(B) 1
(C) 5
(D) 6

78. If A and B are square matrices of order 3 such that $\det.A=-2$ and $\det.B=1$, then $\det.\left(A^{-1}\text{adj}\left(B^{-1}\right).\ \text{adj}\left(2A^{-1}\right)\right)$ is equal to:
(A) 8
(B) –8
(C) 1
(D) –1

79. z_1 and z_2 are two distinct points in an argand plane. If $a|z_1|=b|z_2|$, (where $a,b\in R$) then the point $\frac{az_1}{bz_2}+\frac{bz_2}{az_1}$ is a point on the:
(A) Line segment [- 2, 2] of the real axis
(B) Line segment [- 2, 2] of the imaginary axis
(C) Unit circle $|z|=1$
(D) $z=\tan^{-1}2$

80. A square OABC is formed by line pairs $xy=0$ and $xy+1=x+y$ where 'O' is the origin. A circle with centre C_1 inside the square is drawn to touch the line pair $xy=0$ and another circle with centre C_2 and radius twice that of C_1, is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is:
(A) $\frac{\sqrt{2}}{\sqrt{3}\left(\sqrt{2}+1\right)}$
(B) $\frac{2\sqrt{2}}{3\left(\sqrt{2}+1\right)}$
(C) $\frac{\sqrt{2}}{3\left(\sqrt{2}+1\right)}$
(D) $\frac{\sqrt{2}+1}{3\sqrt{2}}$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let E and M be 3×3 matrices satisfying the system of equations
$EM^T = (EM)^T = 20I$
And $(E+M)^T = 17(E-M)^T$
where I denotes identity matrix of order 3.
If $E^2 + M^2 = \frac{a}{b}I$ (where a and b are co – prime), then find the value of (a + b).

82. Let f(x) be monotonically strictly increasing function in [3, 5] such that $\int_3^5 f^2(x)dx = 9;$
$f(1) = 3; f(4) = 5.$ Find the value of $2\int_1^4 x\left(5 - f^{-1}(x)\right)dx.$

83. Let S_k be the area bounded by the curve $y = x^2(1-x)^k$ and the lines x = 0, y = 0 and x = 1. If $\lim_{n\to\infty}\sum_{k=1}^{n} S_k$ is equal to $\frac{p}{q}$, where p and q are co-prime positive integers, find (p + q).

84. Let $\vec{k}, \vec{l}, \vec{m}, \vec{n}$ are four distinct units vectors in space such that $\vec{k}.\vec{l} = \vec{l}.\vec{m} = \vec{m}.\vec{k} = \vec{n}.\vec{l} = \vec{n}.\vec{m} = \frac{-1}{11}$.
The value of $\vec{k}.\vec{n}$ can be expressed as $\frac{-A}{B}$, where A, B are co-prime positive integer. Find the value of A + B.

85. If eccentricity of conjugate hyperbola of the given hyperbola:
$\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$
is e', then the value of $8e'$ is:

86. Let (a, b) be the outcome of throwing a pair of fair dice. If the probability for which $\lim_{x\to 0}\frac{\ln\left((\cos x)^a\right)}{x^b}$ exist and is finite can be expressed as $\frac{p}{q}$, where p and q are co – prime positive integers, then find the value of (p + q).

87. Let $z_1, z_2 \in C$ and satisfy the equation $|z+1|^2 + |z-1|^2 + 2|z| = 6$. If maximum value of $\left(2|z_1 - 2| + |2z_2 + 1|\right)$ is equal to λ, then find the value of 4λ.

88. Consider $f(x) = x^4 + ax^3 + bx^2 + cx + d$. If straight line $y = 3x + 2$ is tangent to the curve $y = f(x)$ at $P(1, f(1))$ which again intersects $f(x)$ at $Q(2, 8)$ and $f''(1) = 0$, then find the value of $f(3)$.

89. Consider a family of n children. Let two events A and B are defined as follows:
A : is the event that the family has both boys and girls
B : is the event that the family has at most one girl
If the vents A and B are independent, then find the value of n.
[Note: Probability that a randomly selected child is a boy or girl is same]

90. If the function $f(x) = 2x^3 - (8-a)x^2 + \left(a^2 + \frac{16}{9}\right)x - 12$ has local minima at some $x \in R^-$, then find the number of integers in the range of a.